Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

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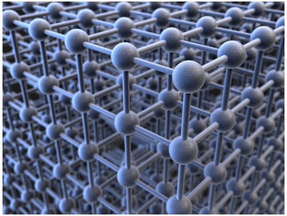
Non-linear phenomena

Crystal Transitions and Disorder

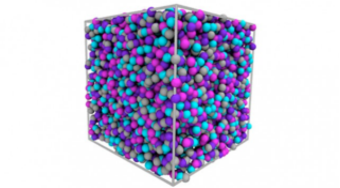
Crystalline



Stronger disorder

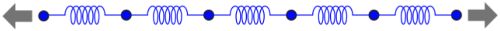
Weak disorder

Amorphous

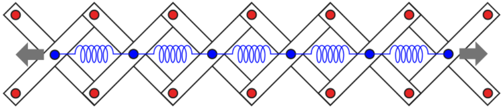


Meta-materials

spring-chain



spring-chain system embedded into a pantograph



Scientific Context: Nonlinearity & Instability

Scientific Challenges

The Challenge:

Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

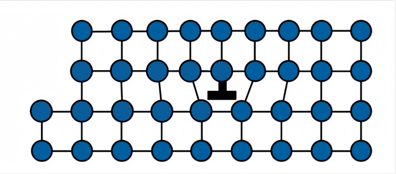
- **Phase Transitions:** Martensitic transformations.
- **Plasticity:** Dislocation-mediated flow and avalanches.
- **Fracture:** Brittle-to-ductile transitions and damage.

Unifying Perspective

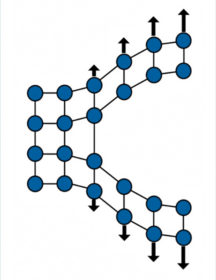
These phenomena are inherently **displacement-driven** (*non-diffusive*) and can be studied within the **same context** of displacive instabilities.

Common Physical Mechanisms: Displacive Instabilities

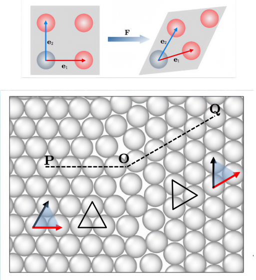
Plasticity



Fracture



Phase Transition

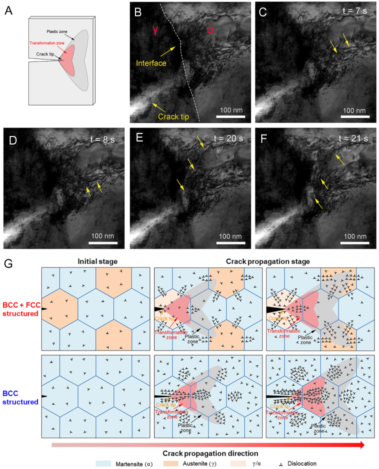


Central Theme

Uniform treatment through the lens of non-convex energy and geometric frustration.

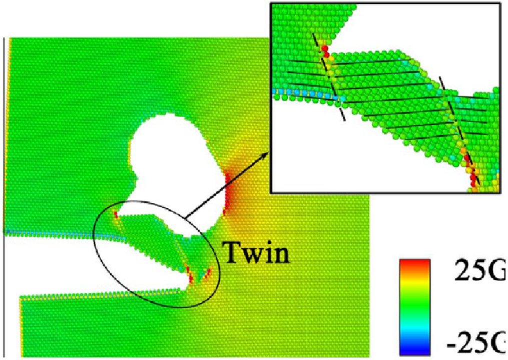
Motivation for Unification

Experimental Observations



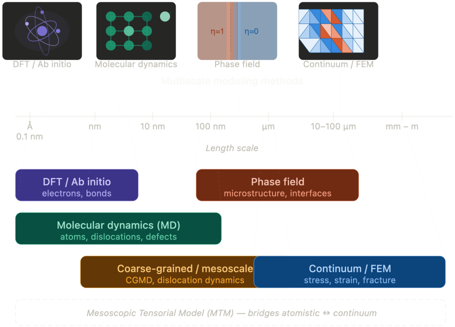
Zhang et al. PNAS, 2025

Computational Modeling



Liu, Groh, Mater. Sci. Eng., 2014

Multiscale Modeling Framework



Bridging Length Scales

Integrating information from the discrete lattice to the continuous engineering scale.